generators. Huge growth of rooftop solar presents business opportunities in rooftop solar system installation.

This project report provides a brief overview of solar rooftop PV plant installation service. Thus it acts as a guide on how to start a business in this area.

Components required

Components of a rooftop solar system are the same anywhere in the world. That said, however, some components make more sense for some regions than others.

For regions such as India and Africa that are keen to use rooftop solar as a source of backup power during grid power outages, batteries could be required at least in select cases. For developed countries such as USA and the European Union where grid outages are rare, batteries are usually not required as backup, with grid supplying the deficit power.

Similarly, some regions of the world require use of trackers for rooftop solar as their rooftop areas are smaller than required to meet their power needs. For some other regions, trackers might not make a good business case.

Key components of a rooftop solar plant include:

Solar photovoltaic modules (panels). There are two kinds of modules, viz, thin-film and crystalline. Rooftop solar plants predominantly use crystalline panels because these are more efficient and therefore better suited to rooftop installations where space is a constraint. Panel efficiency and capacity rating are two important parameters of solar panels.

Inverters. Inverter determines the quality of AC power you get, and also the kind of loads that can be powered with solar. Different inverters support different levels of starting current requirements, which affects the kind of machinery that can be run on solar power. Inverters are also the only major component of the solar plant that are replaced during the lifetime of the plant.

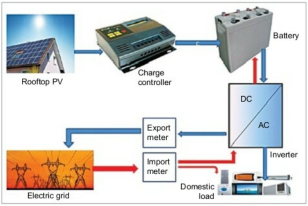

Fig. 2: Block diagram for a grid-connected rooftop PV plant

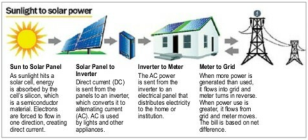

Fig. 3: Process flow for a grid-connected rooftop PV plant

Mounting structures. Solar panels are mounted on iron fixtures so that these can withstand wind and panel weight. The panels are mounted to face south in the northern hemisphere and north in the southern hemisphere for maximum power tracking. Panels are tilted at an angle equal to the latitude of that location.

Proper design of mounting structures is crucial for the power plant performance as the power output from the PV plant cannot be maximised if the mountings buckle and panels are not optimally oriented towards the sun. In addition, improperly mounted panels present a ragged appearance that is not pleasing to the eye. Mounting structures should also allow sufficient air circulation to cool the PV panels as rooftop PV plant output falls with temperatures rising above 25°C.

Solar trackers. Tracking is a mechanism for panel mounting that allows panels to follow the sun as it moves across the sky. Single-axis trackers follow the sun as it moves from east to west during the day, while dual-axis trackers also follow the sun on its north-south journey over the course of a year.

Trackers can increase the power output of a PV plant but add significantly to both the initial cost of the plant and maintenance expenditure. So trackers should be used on a case-to-case basis after performing a cost-benefit analysis over the lifetime of the rooftop plant.

Batteries. A battery pack can add 25-30 per cent to the initial cost of a rooftop PV solar system for one-day autonomy (storing an entire day's output). Inverters with integrated charge controllers are preferred so that the inverter can direct grid power or solar power, based on the availability and demand, to charge batteries. This extends the battery life compared with standalone charge controllers that allow parallel charging between grid and solar power at different power levels, damaging the battery. If batteries are required, it is vital to perform a lifetime cost-benefit analysis to understand the impact on the cost of rooftop solar power.

Other miscellaneous items required include:

1. Power-conditioning unit
2. Array junction box (AJB)/DC distribution box array junction box/DC distribution box
3. Common AC distribution box
4. Common AC distribution panel board
5. Cables; DC cables carry current from panels to inverters, while AC cables carry current from inverters to loads
6. Wire
7. Lightning protection
8. Earthing protection
9. Jet pump

Basics of rooftop solar PV

Solar PV panels (also known as solar PV modules) work by converting sunlight into electricity. These do not use the heat from the sun, and can, in fact, output reduced power in hot climates.